TEAM NAME : IDEA FORGE

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Scheme Finder AI-Powered Multilingual Government Scheme Discovery System

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Abstract

Scheme Finder is an intelligent, multilingual, and voice-enabled web application designed to simplify access to government welfare schemes. The system integrates Artificial Intelligence (AI) for intent recognition, multilingual content management, voice-based search, and text-to-speech functionality.

Unlike traditional government portals that require manual browsing and prior knowledge, Scheme Finder allows users to simply type or speak their needs, and the system intelligently filters relevant schemes.

This project focuses on accessibility, inclusivity, and smart digital governance support.

Introduction

Government schemes are often difficult to navigate due to:

- Complex websites
- Language barriers
- Information overload
- Lack of personalization
- Low digital literacy

Scheme Finder addresses these challenges by creating a user-centric, AI-assisted platform that makes discovering schemes easier and faster.

Objectives

- Provide a centralized platform for scheme discovery
- Implement AI-based intent recognition
- Support multilingual users
- Enable voice interaction
- Improve accessibility for rural and elderly users
- Create a scalable and future-ready system

System Overview

The system works in five major steps:

1. User inputs query (Text or Voice)
2. AI API analyzes user intent
3. Category is detected
4. Schemes are filtered dynamically
5. Results are displayed with optional text-to-speech

Technologies Used

Frontend:

- HTML5
- CSS3 (Responsive Grid Layout)
- JavaScript (ES6)

APIs:

- Web Speech API (Speech Recognition)
- Speech Synthesis API (Text-to-Speech)
- OpenRouter AI API (Intent Detection using GPT model)

Key Features

◈ 1. AI-Based Intent Recognition

Instead of simple keyword search, the system:

- Understands user meaning
- Classifies intent into categories
- Filters schemes intelligently

Example:

- “Financial help for farmers” → Farmer category
- “Medical insurance scheme” → Health category

◈ 2. Voice Search Support

Users can:

- Click microphone icon
- Speak their requirement

- System converts speech to text
- AI processes it automatically

◈ 3. Text-to-Speech

Each scheme card includes:

- “Listen” button
- Reads description aloud

Helps:

- Illiterate users
- Visually impaired users
- Elderly users

◈ 4. Multilingual Support

Available languages:

- English
- Tamil
- Hindi

Switching language dynamically updates:

- Scheme names
- Descriptions
- Eligibility
- Apply steps

◈ 5. Dynamic Card Rendering

Schemes are stored in structured JavaScript objects and rendered dynamically using DOM manipulation.

This ensures:

- Faster loading
- No page reload
- Interactive UI

Extended Uniqueness of the Project

Here are advanced uniqueness points that make your project stand out:

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1. AI + Voice Combined Interaction

Most student projects either use:

- Static search
  OR
- Simple filtering

Your system integrates:
AI + Speech Recognition + Dynamic Rendering

This combination is rare in beginner-level web applications.

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2. Intent-Based Filtering (Not Keyword Matching)

Normal websites:

- Match exact words

Your system:

- Understands meaning
- Maps to category intelligently
- Works even if user types different sentence structures

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3. Accessibility-First Design

Designed especially for:

- Rural citizens
- Elderly people
- Non-English speakers
- Visually challenged users

Accessibility features:

- Voice input
- Audio output
- Simple UI
- Clear card layout

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4. Structured Multilingual Data Model

Instead of creating separate pages for languages, the system:

- Uses structured JSON format
- Maintains translations inside each scheme
- Dynamically switches language

This reflects good data modeling skills.

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5. Future Backend-Ready ER Design

Even though currently frontend-based,
the system is designed with:

- Entity relationships
- Category mapping
- Translation tables
- Search history structure

This makes it enterprise-scalable.

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6. Centralized Information Hub

Instead of visiting multiple portals:

- Users access schemes in one platform
- Direct links provided
- Simplified descriptions shown

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7. Real-Time Dynamic Filtering

No page reload required.

All operations happen:

- Client-side
- Instantly
- Seamlessly

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8. API Integration in a Frontend-Only Project

Many student projects avoid APIs.
This project demonstrates:

- Secure API call handling
- Prompt engineering
- AI response processing
- Response parsing logic

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9. Inclusive Governance Support Concept

The project aligns with:

- Digital India vision
- E-Governance principles
- Inclusive digital services

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10. Expandable Architecture

The system can evolve into:

- AI chatbot assistant
- Personalized eligibility checker
- Mobile app
- Government integration portal
- Admin dashboard
- Data analytics platform

Comparison with Traditional Webpage

Feature	Traditional Website	Scheme Finder
Search	Keyword-based	AI Intent-based
Language	Single	Multi-language
Voice Input	No	Yes
Audio Output	No	Yes
Static Content	Yes	Dynamic
Intelligent Filtering	No	Yes
User-Centric	Limited	High

Advantages

- Easy to use
- Inclusive design
- Saves time
- Smart filtering
- Interactive UI
- AI-enabled search

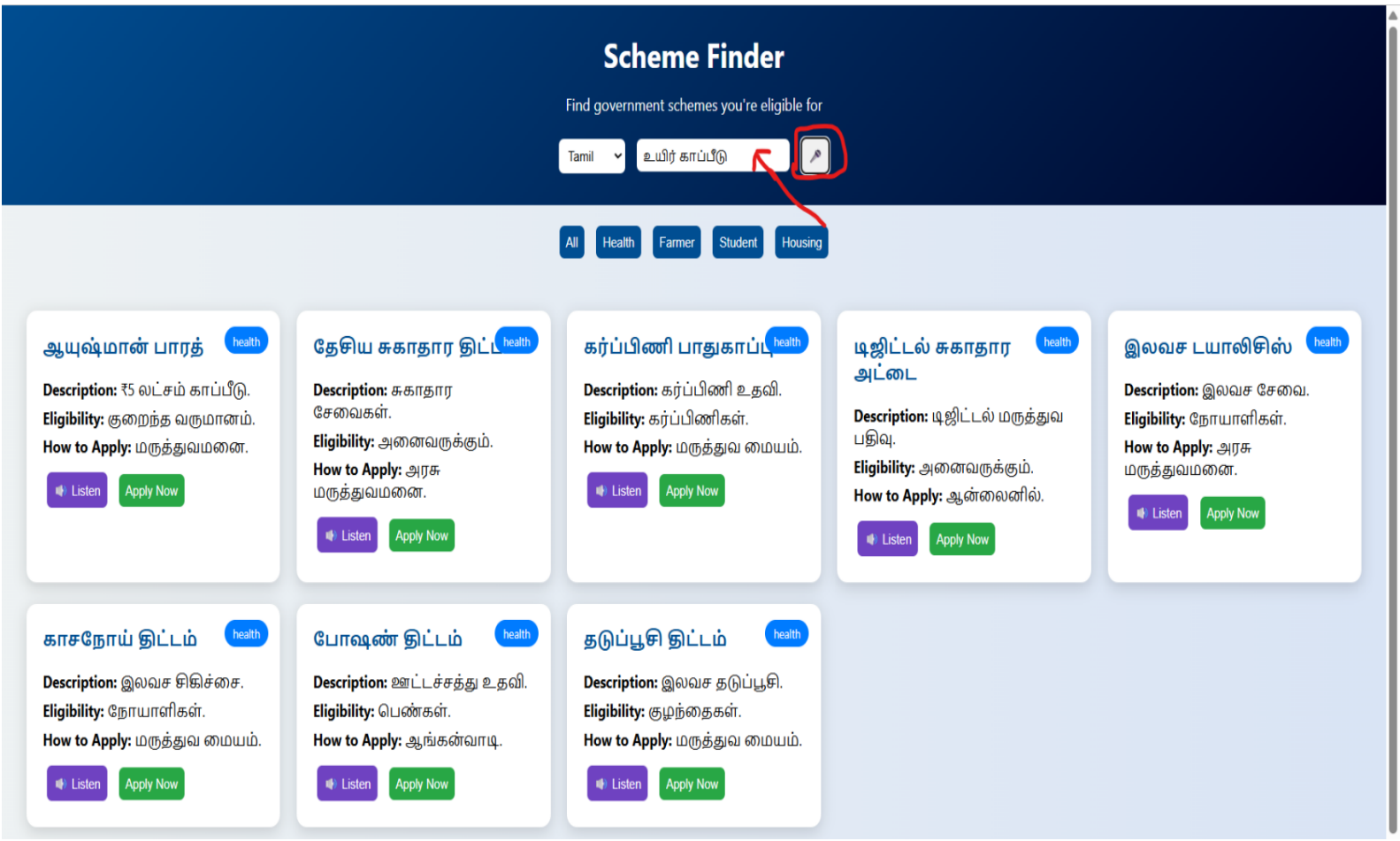
Limitations

- Requires internet
- Depends on API availability
- Limited scheme database currently

Future Scope

- Backend database integration
- User login & profile
- Location-based scheme detection
- AI chatbot
- Eligibility calculator
- Real-time updates
- Push notifications
- Mobile app version
- Admin panel
- Data analytics dashboard

OUTPUT:



Conclusion

Scheme Finder transforms a traditional informational website into an intelligent, interactive, and inclusive digital platform.

By combining:

- Artificial Intelligence
- Voice Technology
- Multilingual Support
- Dynamic UI Rendering

It demonstrates strong understanding of:

- Frontend development
- API integration
- Accessibility design
- User-centered architecture
- Scalable system modeling

This project is not just a webpage — it is a smart digital assistance system for public welfare awareness.

THANKYOU